Question			Answer	Marks	AO	Guidance
10	(a)		When $t = 0, A = 8$ so $a = 8$ When $t = 1, A = 8.8$ so $b = 0.8$	B1 B1 [2]	3.3 3.3	Allow embedded in $A = 8 + 0.8t$
10	(b)		$15 = 8 + 0.8t$ so $t = 8.75$	B1 [1]	3.4	Allow for "after 9 days" oe FT their values in (a)
10	(c)		The model is linear so gives the same increase each day. 10% increase each day gives bigger increases as the size of the culture increases.	B1 [1]	3.5b	Must indicate the mis-match between the linear model and the exponential observed results either in general terms or for a particular day (Amounts are 8, 8.8 and 9.6 for the model and 8, 8.8 9.68 for the 10% increase) (see appendix)
10	(d)		Using $A = Pe^{kt}$ when $t = 0$ gives $P = 8$ When $t = 1$, $8.8 = 8e^{1k}$ So $k = \ln 1.1 = [0.0953...]$	B1 M1 A1 [3]	3.3 3.3 3.3	cao Forming an equation for k using $t = 1$ and $A = 8.8$ oe FT their value for P Allow for $\ln 1.1$ or a decimal answer to at least 2sf
10	(e)		Area 15 cm^2 when $15 = 8e^{(\ln 1.1)t}$ $t = \dfrac{\ln\left(\frac{15}{8}\right)}{\ln 1.1} = 6.60$	M1 A1 [2]	3.4 1.1	Correct use of logs in an attempt to solve indicial equation Similarly for $15 = 8 \times 1.1^t$ FT their P and k
10	(f)		The model predicts unlimited growth which is not possible in the laboratory	B1 [1]	3.5b	Must describe what the model predicts and compare with the situation being modelled (see appendix)

Exemplar responses for Q10(c)

Response	Mark
Larger values of t give inaccurate results	**B0**
Exponential doesn't give the same increase each day	**B0**
10% of the new area is not 10% of the original area	**B0**
Original model predicts an increase of 0.8 each day but the increase 10% each day so the model is an underestimate	**B1**
This model doesn't grow exponentially but increasing by 10% each day would be modelled by that	**B1**
The model is linear whereas 10% each day is exponential	**B1**

Exemplar responses for Q10(f)

Response	Mark
It would get too big	**B0**
It would get impossibly big	**B1**
There is no limit to the size of the area, but growth will eventually reduce or stop	**B0**
The area *would be* too large to be plausible and have to outgrow the lab (would be implies the model)	**B1 BOD**
For large values of t, the area would be too large (large values of t implies the model being used)	**B0**
For large values of t, the area would be impossibly large	**B1**
It suggests that the culture grows to a size that is unrealistic ("it" refers to the model)	**B1**
The model predicts unlimited growth but this is not possible	**B1**